

Homework 6: Logic Coverage of Thermostat Predicate

The predicate tested in this homework is:

$(\text{curTemp} < \text{targetTemp} - \text{thresholdDiff}) \ \&\& \ (\text{timeSinceLastRun} \geq \text{minLag})$

Define:

$A = (\text{curTemp} < \text{targetTemp} - \text{thresholdDiff})$ $B = (\text{timeSinceLastRun} \geq \text{minLag})$ $P = A \ \&\& \ B$

Predicate Coverage (PC)

Predicate Coverage requires the predicate to evaluate to both TRUE and FALSE.

Test Case A B Predicate Result T1 True True True T2 False True False

PC is satisfied because the predicate evaluates to both TRUE and FALSE.

Clause Coverage (CC)

Clause Coverage requires every clause to evaluate to both TRUE and FALSE.

Test Case A B T1 True True T2 False True T3 True False T4 False False

CC is satisfied because:

Clause A evaluates to both TRUE and FALSE. Clause B evaluates to both TRUE and FALSE. Correlated Active Clause Coverage (CACC)

CACC requires each clause to independently affect the predicate outcome.

Major Clause A Test Case A B Predicate T1 True True True T2 False True False

B remains fixed while A changes the predicate result.

Major Clause B Test Case A B Predicate T1 True True True T3 True False False

A remains fixed while B changes the predicate result.

Therefore, CACC is satisfied.

Test Execution Result

All test cases passed successfully.

T1 PASS T2 PASS T3 PASS T4 PASS

Conclusion

This homework successfully designed test cases for:

Predicate Coverage (PC) Clause Coverage (CC) Correlated Active Clause Coverage (CACC)

All logical coverage requirements were satisfied for the Thermostat class.

GitHub Repository

<https://github.com/tainyu/secure-software-testing-2025-spring-classroom-lab-2-114-spring-secure-software-testing-1/tree/main/HW6>